

Abstract No.

CDKN2A loss defines a MAPK/RB signaling dependency in BRAF-wildtype GBM and CDK4/6+MEK dual inhibition overcome adaptive resistance

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Glioblastoma (GBM) remains incurable despite aggressive therapeutic approaches. The MAPK pathway is a conserved, MGMT methylation-agnostic, and subtype-independent feature of GBM. In oncology, MAPK inhibition has been clinically restricted, in combination with dabrafenib, to BRAF-mutant patients, yet responses have remained transient because of adaptive resistance mechanisms. In GBM, clinical evaluation of MAPK inhibition has largely been confined to the rare subset of BRAF-mutant tumors (<5%), leaving its therapeutic potential in BRAF-wt disease largely unexplored. By integrating functional screening across a molecularly heterogeneous panel of patient-derived GBM stem-like cell (GSC; n=10; IDH-wt, BRAF-wt) with patient-derived orthotopic xenograft (PDX), we investigated the efficacy of MEK inhibition using trametinib.

Trametinib induced dose-dependent suppression of proliferation and self-renewal in most GSCs, whereas a subset exhibited reduced sensitivity. Intrinsic trametinib sensitivity perfectly segregated with CDKN2A status: all sensitive GSC models harbored CDKN2A-deletion, whereas all resistant models retained intact-CDKN2A.

Mechanistic studies included cell-cycle analysis, 3D gliomasphere assays, protein profiling, LC-MS/MS drug quantification, and loss-of-function validation using shRNA. Upon MEK inhibition, CDKN2A-deleted GSCs underwent robust Rb-dependent G1 arrest. In contrast, CDKN2A-intact cells failed to establish effective trametinib-dependent cell-cycle blockade. Silencing CDKN2A in resistant, Rb-intact GSCs was sufficient to enhance trametinib sensitivity. In CDKN2A-deleted PDX model, a ten-days trametinib treatment delayed tumor growth and extended survival by approximately 19 days (p<0.05, +17.4%). Prolonged trametinib exposure in-vitro induced adaptive resistance characterized by Rb reactivation and increased AKT signaling. In-vivo, combining trametinib with the brain-penetrant CDK4/6 inhibitor abemaciclib effectively overcame adaptive resistance, significantly extending survival compared with monotherapy (p<0.0001, +41.5%). These findings identify CDKN2A-loss as predictive biomarker of MEK-inhibition response in BRAF-wt GBM, revealing a previously unrecognized MAPK/Rb signaling vulnerability. Adaptive resistance can be overcome through combined CDK4/6 and MEK inhibition, providing a rationale for a new strategy in CDKN2A-deleted BRAF-wt GBM.

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